they are, much less where they are. But there is at least a chance that sometime in the remote future one of the *Voyagers* will be intercepted and examined by an alien craft.

Accordingly, as each *Voyager* left Earth for the planets and the stars, it carried with it a golden phonograph record encased in a golden, mirrored jacket containing, among other things: greetings in 59 human languages and one whale language; a 12-minute sound essay including a kiss, a baby's cry, and an EEG record of the meditations of a young woman in love; 116 encoded pictures, on our science, our civilization, and ourselves; and 90 minutes of the Earth's greatest hits—Eastern and Western, classical and folk, including a Navajo night chant, a Japanese *shakuhachi* piece, a Pygmy girl's initiation song, a Peruvian wedding song, a 3,000-year-old composition for the *ch'in* called "Flowing Streams," Bach, Beethoven, Mozart, Stravinsky, Louis Armstrong, Blind Willie Johnson, and Chuck Berry's "Johnny B. Goode."

Space is nearly empty. There is virtually no chance that one of the *Voyagers* will ever enter another solar system—and this is true even if every star in the sky is accompanied by planets. The instructions on the record jackets, written in what we believe to be readily comprehensible scientific hieroglyphics, can be read, and the contents of the records understood, only if alien beings, somewhere in the remote future, find *Voyager* in the depths of interstellar space. Since both *Voyagers* will circle the center of the Milky Way Galaxy essentially forever, there is plenty of time for the records to be found—if there's anyone out there to do the finding.

We cannot know how much of the records they would understand. Surely the greetings will be incomprehensible, but their intent may not be. (We thought it would be impolite not to say hello.) The hypothetical aliens are bound to be very different from us—independently evolved on another world. Are we really sure they could understand anything at all of our message? Every time I feel these concerns stirring, though, I reassure myself: Whatever the incomprehensibilities of the *Voyager* record, any alien ship that finds it will have another standard by which to judge us. Each *Voyager* is itself a message. In their exploratory intent, in the lofty ambition of